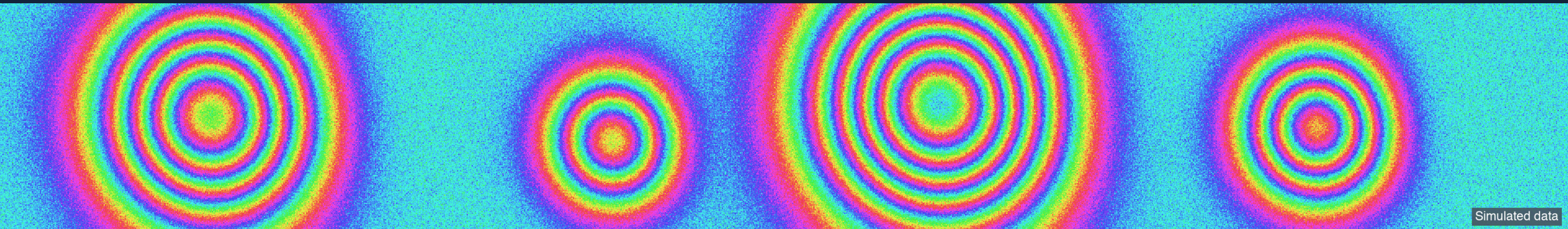
ENVIRONMENTAL GEOHAZARDS ENGINEERING · LECTURE

Land Subsidence Monitoring with InSAR

Measuring sinking ground in cities, farmland and coasts

AKSHAY RAJ MANOCHA



GOAL OF THIS LECTURE

Use satellite radar to find where the ground is sinking, explain why, and judge what it means for buildings, infrastructure and flooding.

Causes and mechanics, measuring vertical motion, linking it to groundwater, and real cities from Mexico City to Delhi.

THIS LECTURE

Outline

01 **What land subsidence is and why it happens**
Causes, aquifer mechanics, and what InSAR adds

02 **Measuring subsidence with InSAR**
Vertical motion, two passes, techniques and ground checks

03 **Reading the results**
Subsidence bowls, time series, groundwater and differential settlement

04 **Case studies**
Mexico City, Jakarta, Delhi, California’s Central Valley and the global picture

05 **Engineering practice**
Impacts, management, limits and exercises

01

PART 1

What land subsidence is and why it happens

Causes, aquifer mechanics, and what InSAR adds

SUBSIDENCE

The ground is sinking

Land subsidence is the gradual lowering of the ground surface. It is usually millimetres to centimetres a year, adds up over decades, and is often permanent.

≈ 40 cm/yr
Fastest sinking measured in Mexico City with Sentinel-1

9 m
Sinking near Mendota, California, from 1925 to 1977

19%
Of the world’s population may face a high probability of subsidence

635 million
People in flood-prone subsiding areas by 2040 (projection)

Sources: Cigna and Tapete (2021); USGS; Herrera-García et al. (2021), Science.

SUBSIDENCE

Why the ground sinks

Groundwater pumping

The most common cause: clay layers in the aquifer compact as water is removed.

Oil and gas extraction

Reservoir rocks compact as fluids are produced; coastal oil fields have sunk by metres.

Mining

Underground voids collapse above coal and salt mines, as in India’s Jharia coalfield.

Drained organic soils

Peat shrinks and oxidises when drained for farming, in deltas and wetlands.

Natural and load compaction

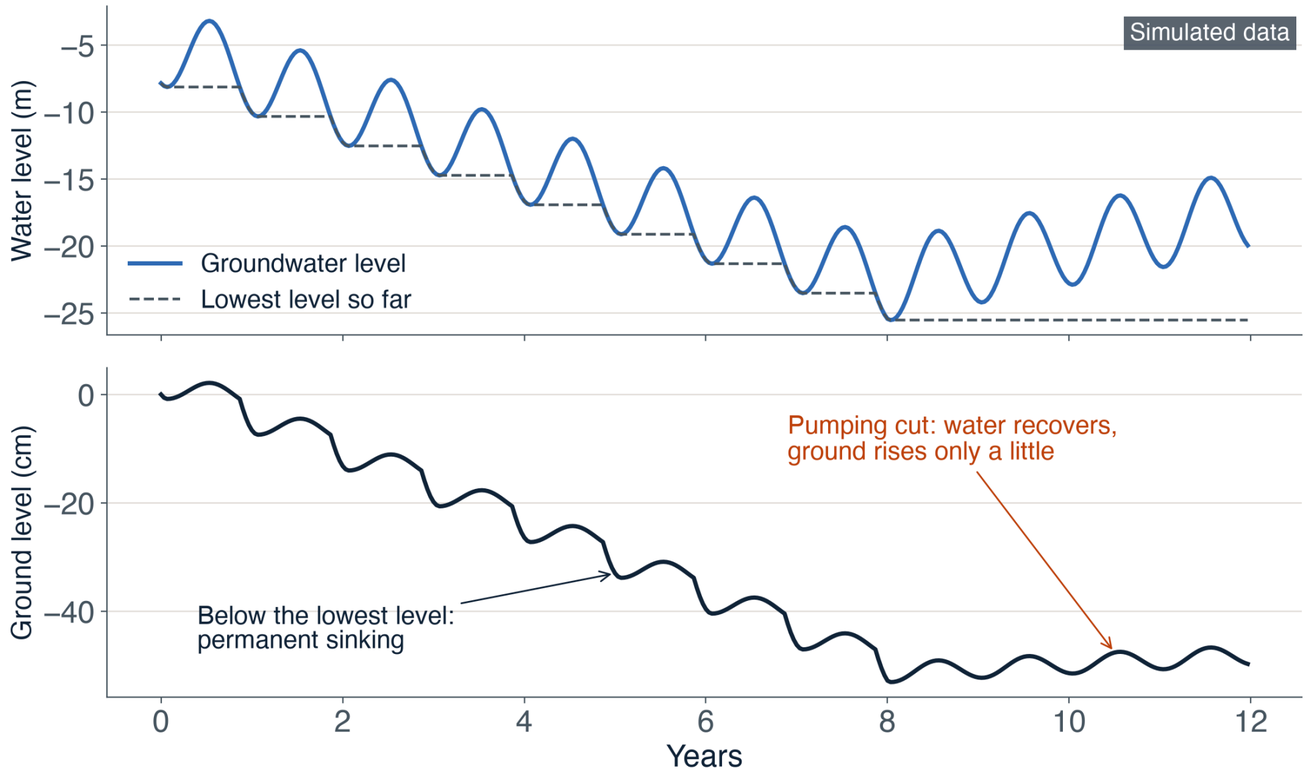
Young delta sediments, reclaimed land and heavy new buildings settle.

Dissolution and sinkholes

Limestone, gypsum and salt dissolve; the ground above can collapse suddenly.

SUBSIDENCE

How pumping makes the ground sink



Simulated aquifer system: elastic and permanent compaction.

Lower water pressure means the grains carry more of the weight above (effective stress).

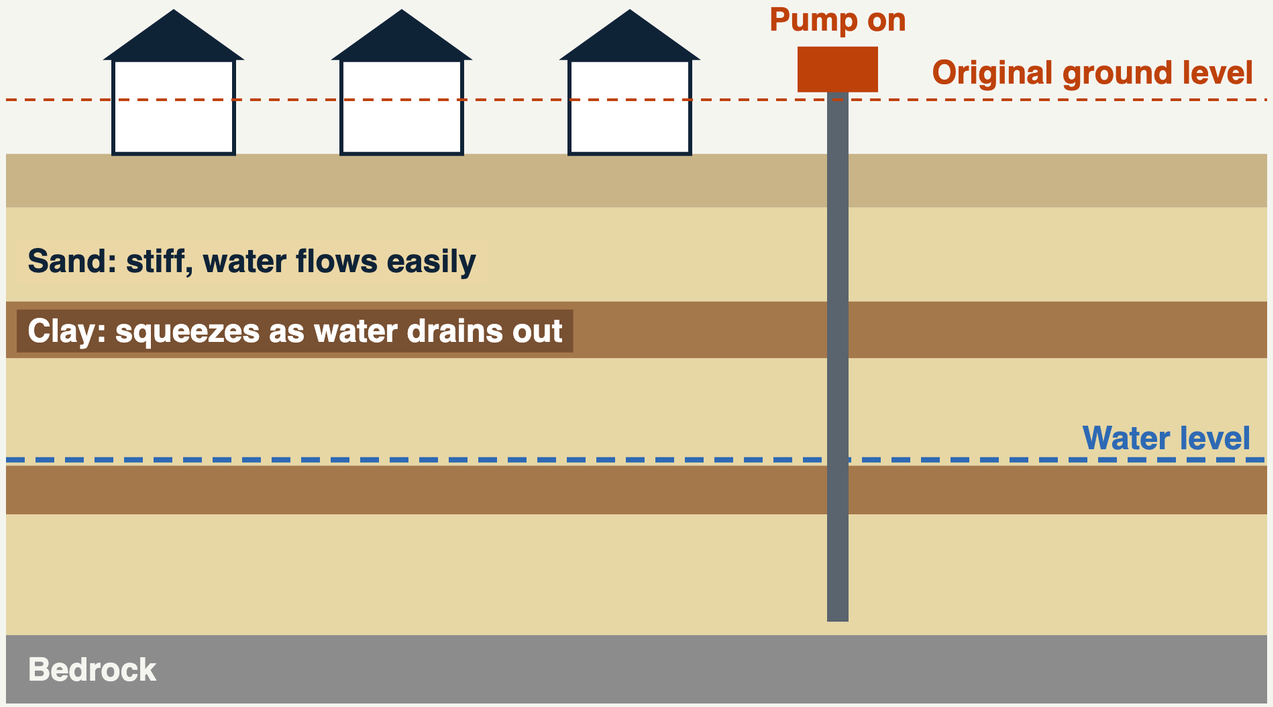
Sand compresses a little and springs back: elastic.

Clay compresses a lot once the water falls below its previous lowest level, and stays compressed: permanent.

Thick clays keep compacting for years after water levels stop falling.

SUBSIDENCE

Pump, squeeze, sink



Pumping:
water falls, clay squeezes, ground sinks

Water level



Ground level



Most of the sinking is permanent

SUBSIDENCE

Why InSAR is the right tool

mm/yr
Precision of a good time series over a city

250 km
Width of one Sentinel-1 image: a whole city or basin

Since 1992
Archive: ERS and Envisat, then Sentinel-1 from 2014

Millions
Measurement points over a large city

Finds the sinking areas

Bowls centred on pumping areas show up even where nobody suspected them.

Measures every building

Persistent scatterers on roofs and walls give a value for individual structures.

Checks water policy

Did pumping limits or recharge work? The next year of images will show it.

02

PART 2

Measuring subsidence with InSAR

Vertical motion, two passes, techniques and ground checks

MEASURING

Mostly vertical, but not only

$$v(up) \approx v(LOS) \div \cos \theta$$

If the ground only moves up or down, the line of sight sees a share cos θ of it: about 78% for Sentinel-1 at θ = 39°.

Example

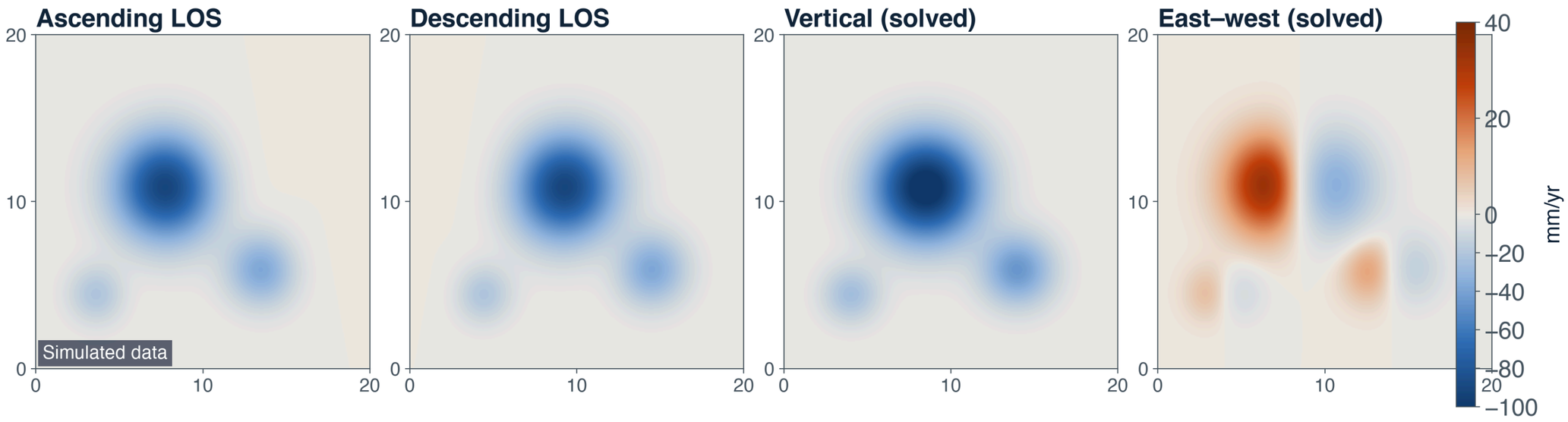
LOS velocity −39 mm/yr at the centre of a bowl: vertical ≈ −39 ÷ 0.78 ≈ **−50 mm/yr**.

But on the flanks

The ground also moves sideways, towards the centre of the bowl. Ascending and descending then disagree, and the simple division is wrong. Use both passes.

MEASURING

Two passes: vertical and east–west



Two looks, two unknowns. East–west: red = moving east, blue = west (towards the bowl centre).

Simulated data. Solving two equations for two unknowns recovers the vertical motion to within about 5 mm/yr; north–south motion is ignored.

Vertical: the subsidence itself, ready for engineering use.

East–west: sideways motion towards each bowl, which strains pipes and stretches buildings.

MEASURING

Techniques and ground checks

Method	Best for	Limit
Persistent scatterers (PS-InSAR)	Cities: buildings, roads, bridges; mm/yr	Few points in fields
Small baseline (SBAS)	Farmland and suburbs: whole basins	Coarser; seasonal crops break coherence
Ready-made products	EGMS (Europe), COMET-LiCSAR, national services	Fixed processing choices
Levelling and GNSS	Calibrate and check InSAR; a stable reference	Few points
Borehole extensometers	Which layer is compacting, and how fast	One site per borehole
Groundwater wells	The cause: water levels to compare with the sinking	Uneven coverage

03

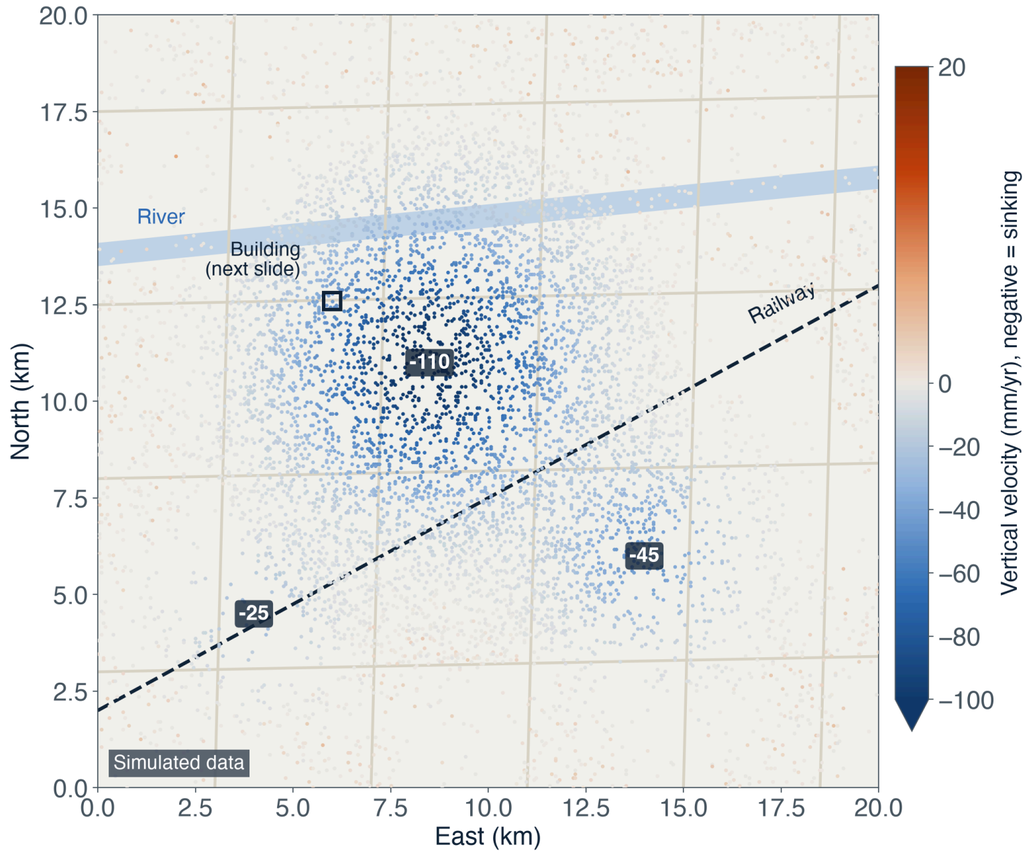
PART 3

Reading the results

Subsidence bowls, time series, groundwater and differential settlement

RESULTS

Subsidence bowls in a city

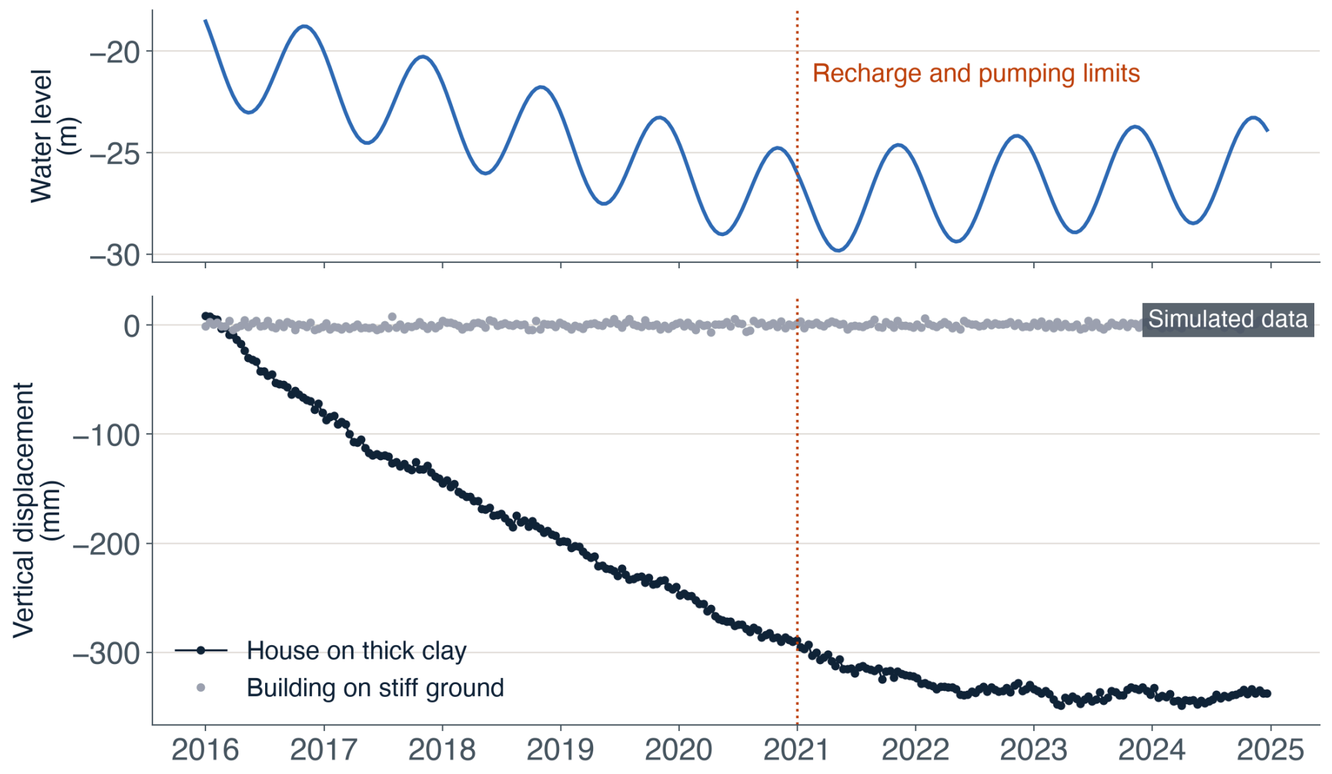


Simulated data: vertical velocity from both passes.

- Bowls form over areas of heavy pumping: industry, dense housing, irrigated fields.
- Dense points in the built-up core; few in fields, where crops break coherence.
- The railway crosses the edge of a bowl: one end sinks faster than the other.
- The river side sinks too: levees and drains lose height.

RESULTS

Time series: the ground follows the water



Simulated data: Sentinel-1 every 12 days.

On thick clay the ground sank about 58 mm/yr while the water level fell.

Stiff ground nearby barely moves: the difference is the geology, not the city.

After recharge and pumping limits, sinking slowed to about 7 mm/yr within two years.

Clays respond with a delay: expect a lag of months to years.

RESULTS

From sinking ground to lost groundwater

$$S = \Delta b \div \Delta h$$

Skeletal storage: compaction Δb (from InSAR) per metre of water-level change Δh (from wells).

≈ 2 mm/m

Elastic, from the seasonal swing: recoverable

≈ 30 mm/m

Permanent, from the long-term decline: lost for good

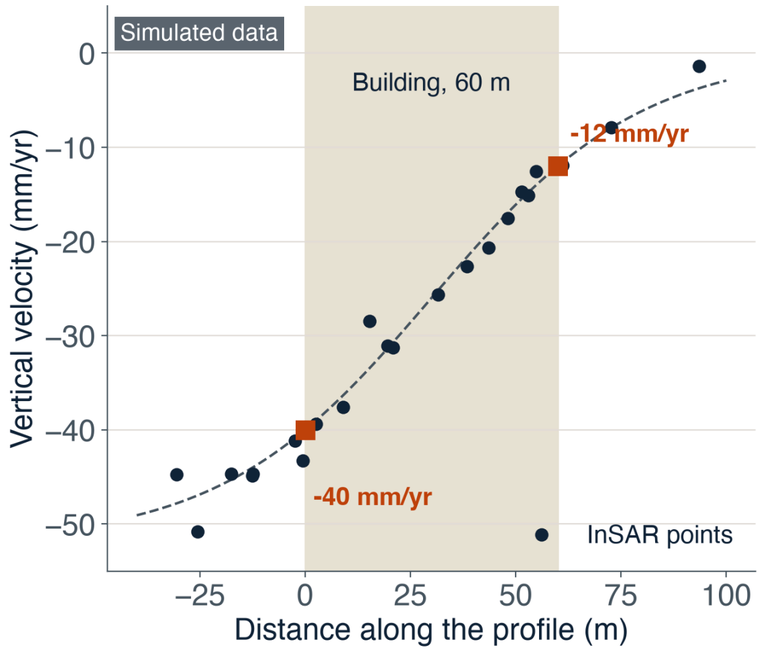
InSAR plus well data measure how much of the aquifer’s capacity has been lost.

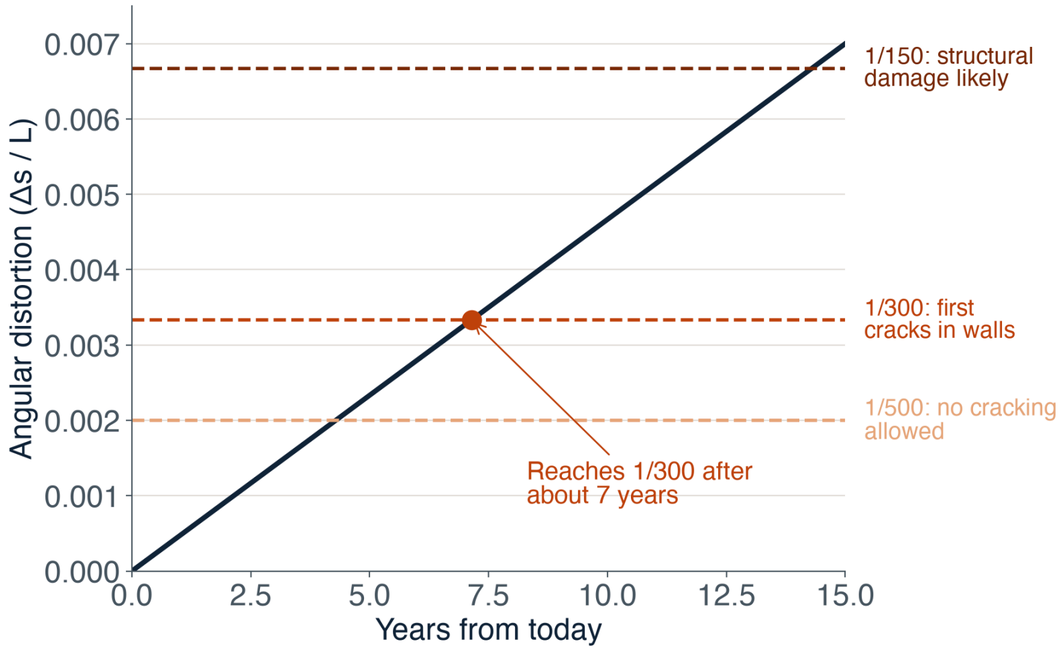
The same numbers predict how much more the ground will sink if levels keep falling.

Values here are from the simulated time series on the previous slide.

RESULTS

What damages buildings: differential settlement





Simulated data. Angular distortion = difference in settlement ÷ distance. Damage limits after Bjerrum (1963).

Here: 28 mm/yr across 60 m reaches 1/500 in about 4 years and 1/300 in about 7.

Lesson: a building that sinks evenly is safe; one that sinks unevenly cracks.

04

PART 4

Case studies

Mexico City, Jakarta, Delhi, California’s Central Valley and the global picture

CASE STUDY

Mexico City, Mexico

≈ 42 cm/yr
Fastest sinking, north-east of the city, Sentinel-1 2014–2020

100+ years
Of continuous sinking from groundwater pumping

Lake clay
The city stands on the soft clays of a drained lake

Faults
Subsidence opens ground cracks that damage buildings and pipes

- Sinking is fastest where the lake clays are thickest.
- InSAR maps where buildings, metro lines and water pipes cross steep changes in rate.
- Most of the lost height and aquifer storage will not come back.

Sources: Cigna and Tapete (2021), Remote Sensing of Environment; Chaussard et al. (2021), JGR Solid Earth.

CASE STUDY

Jakarta, Indonesia

1–15 cm/yr
Typical sinking rates across the city

20–28 cm/yr
Fastest rates, at some places and times

Coast
Sinking plus sea-level rise: more frequent floods

2022
Law to move the capital to Nusantara, on Borneo

Groundwater pumping, where piped water is scarce, is the main cause.

North Jakarta, by the sea, sinks fastest; sea walls are being raised.

Rates change over short distances: point-by-point InSAR is needed.

Sources: Abidin et al. (2011), Natural Hazards; Proc. IAHS 372 (2015); InSAR studies of the Jakarta basin.

CASE STUDY

Delhi National Capital Region, India

> 15 cm/yr
Fastest sinking, near Indira Gandhi International Airport

4–5 cm/yr
Sinking in Faridabad and Gurugram

≈ 2 cm/yr
Uplift in Dwarka after recharge: the aquifer recovering

15 → 7 cm/yr
Central sinking slowed, 2014–18 to 2019–23

- Groundwater over-use is the main driver of the sinking.
- Recharge structures and pumping limits stopped sinking in Dwarka.
- InSAR showed both the problem and whether the policy worked.

Sources: Garg et al. (2022), Scientific Reports; Kumar et al. (2025), Water Resources Research.

CASE STUDY

San Joaquin Valley, California

9 m
Sinking near Mendota from 1925 to 1977

33 cm
Sinking near Corcoran in 8 months of the 2015 drought

30 m
Groundwater below previous record lows in the drought

2 bowls
Centred on Corcoran and south of El Nido

Irrigation pumping peaks in droughts, and so does the sinking.

Canals, aqueducts, bridges and wells are damaged; canals lose capacity.

California’s 2014 groundwater law treats subsidence as an outcome to avoid; InSAR tracks it.

Sources: USGS (J. F. Poland); NASA JPL (2015) report on Central Valley subsidence.

CASE STUDY

Sinking cities worldwide

19%
Of the world’s population may face a high probability of subsidence

878 km²
Sinking land in five Indian megacities, 2015–23

2,406
Buildings at high risk of damage now in Delhi, Mumbai and Chennai

24 cm/yr
Tokyo in 1968, before pumping limits stopped it

Groundwater-driven subsidence is growing with cities and droughts.

In India, about 1.9 million people live on ground sinking more than 4 mm/yr.

Tokyo and Osaka show the cure: regulate pumping, supply surface water.

Sources: Herrera-García et al. (2021), Science; Nature Sustainability (2025), ‘Building damage risk in sinking Indian megacities’; IGES (Tokyo).

05

PART 5

Engineering practice

Impacts, management, limits and exercises

ENGINEERING

What subsidence does to infrastructure

Buildings

Uneven settlement tilts and cracks walls and frames.

Pipes and sewers

Joints break; gravity sewers lose their slope and back up.

Flooding

Low ground floods more often; levees and sea walls lose height.

Transport

Rail track, roads, bridge joints and runways go out of level.

Water supply

Canals lose capacity, well casings fail, the aquifer stores less.

Ground cracks

Earth fissures open at the edges of bowls and over buried steps.

ENGINEERING

Slowing and stopping subsidence

Limit pumping

Licences and limits on wells, as in Tokyo and Osaka from the 1960s.

Recharge the aquifer

Recharge wells and ponds raise water levels, as in Dwarka, Delhi.

Supply surface water

Piped water from rivers and reservoirs replaces groundwater.

Monitor and plan

InSAR, GNSS and extensometers every year; zoning and design for settlement.

ENGINEERING

Limits and good practice

Limits

- Line of sight mixes vertical and horizontal motion
- Few points in fields and forests; crops break coherence
- Needs a stable reference area
- Shows the surface, not which layer is compacting

Good practice

- Use both passes; solve for vertical
- Calibrate with GNSS and levelling
- Compare with wells and borehole extensometers
- Report differential settlement, not just rates

EXERCISES

Try these

01 A Sentinel-1 pass (incidence 39°) measures −39 mm/yr at the centre of a bowl. If the motion is vertical, what is the sinking rate?

02 The water level falls 1.5 m a year and a clay site sinks 45 mm a year. How many mm of compaction per metre of water-level fall is that?

03 A 60 m building sinks 40 mm/yr at one end and 12 mm/yr at the other. What is its angular distortion after 5 years? Is it past 1/500?

04 A coastal city sinks 25 mm/yr while the sea rises 4 mm/yr. How much higher does the sea get relative to the land in 20 years?

EXERCISES

Answers

01 −39 ÷ 0.78 ≈ **−50 mm/yr** (sinking).

02 45 mm ÷ 1.5 m = **30 mm per metre**: mostly permanent, inelastic compaction.

03 (40 − 12) × 5 = 140 mm over 60 m: 0.14 ÷ 60 ≈ **1/429**. Yes, past 1/500 (no cracking allowed), not yet 1/300.

04 (25 + 4) × 20 = 580 mm: the sea is about **0.6 m** higher relative to the land. Here sinking matters six times more than sea-level rise.

RECAP

Key points

01 Most subsidence today comes from pumping groundwater out of clay-rich aquifers.

02 Clay compaction is slow, delayed and mostly permanent.

03 InSAR measures it everywhere; two passes give the vertical motion.

04 Differential settlement, not total sinking, damages buildings and pipes.

05 Managing groundwater can stop it, and InSAR shows whether it worked.

FURTHER READING

References

Abidin, H. Z. et al. (2011). Land subsidence of Jakarta (Indonesia) and its relation with urban development. Natural Hazards 59, 1753–1771.

Bjerrum, L. (1963). Allowable settlement of structures. Proc. European Conference on Soil Mechanics and Foundation Engineering, Wiesbaden, vol. 2.

Chaussard, E. et al. (2021). Over a century of sinking in Mexico City: no hope for significant elevation and storage capacity recovery. JGR Solid Earth 126.

Cigna, F., Tapete, D. (2021). Present-day land subsidence rates, surface faulting hazard and risk in Mexico City with 2014–2020 Sentinel-1 IW InSAR. Remote Sensing of Environment.

Galloway, D. L., Burbey, T. J. (2011). Review: regional land subsidence accompanying groundwater extraction. Hydrogeology Journal 19, 1459–1486.

Garg, S., Motagh, M., Indu, J., Karanam, V. (2022). Tracking hidden crisis in India’s capital from space: implications of unsustainable groundwater use. Scientific Reports 12.

Herrera-García, G. et al. (2021). Mapping the global threat of land subsidence. Science 371, 34–36.

Kumar et al. (2025). InSAR reveals recovery of stressed aquifer systems in parts of Delhi, India: evidence for improved groundwater management. Water Resources Research.

Nature Sustainability (2025). Building damage risk in sinking Indian megacities.

Poland, J. F. et al. (1975). Land subsidence in the San Joaquin Valley, California, as of 1972. USGS Professional Paper 437-H.

Thank you. Questions?

Figures and the animation were made for this lecture. Maps, time series and the settlement example are simulated with Sentinel-1 geometry, and labelled as such. Damage limits after Bjerrum (1963). Case-study facts: Cigna and Tapete (2021) and Chaussard et al. (2021) (Mexico City); Abidin et al. (Jakarta); Garg et al. (2022) and Kumar et al. (2025) (Delhi); USGS and NASA JPL (California); Herrera-García et al. (2021), Nature Sustainability (2025) and IGES (global). Sentinel-1 data: EU Copernicus programme / ESA. Full references on the previous slide.